

Problem Statement 2

Background

Inter-IIT High and Mid Prep PSs have generally been more research-based in recent years. There have also been some problem statements that are more technical and implementation-based, which are covered by the other practice problem statements given to you.

One major problem, especially with 2nd-year students (and some 3rd year as well), is the ability to read research papers, distinguish bad papers from good ones, and search for proper papers related to a required problem statement. Since 2nd-year students generally do not have much experience with research-based problems, this PS is designed to check this skill.

Question

Some of the most exciting research in the AI and robotics area is the development of **general-purpose robotics**. Two major ideas in this area are **Vision-Language-Action (VLA) models** and **World Action Models (WAMs)**. Some companies, such as Google DeepMind and Physical Intelligence, are still betting heavily on VLA models, while others, such as NVIDIA, are shifting towards WAMs. Many researchers are also suggesting a hybrid approach combining these ideas.

You are requested to study some papers in this area and give your opinion on **which approach you think is more suitable and why**.

Refer to survey papers as well. Read **3 research papers** related to this topic. Make sure that the three papers are **not on very similar topics**. You should explore different aspects of the problem.

Make proper notes for each paper. We may ask questions based on these papers during the evaluation.

If a topic is too advanced (for example, if it requires concepts such as transformers that you do not yet understand) or is too mathematical, you can skip those parts. The main thing we are checking is **whether you can research a problem properly and form your own understanding and opinion**. This skill will be vital for any High or Mid Prep PS.

If you are interested in robotics, this reading will also be very useful for your personal growth and understanding of the field.

LLM Usage Guidelines

You are free to use an LLM as an assistant. However, **do not simply copy-paste the entire question into a Pro model and submit the generated answer**.

Try to form your **own opinion** about the topic. Your opinion may be wrong—that is completely fine, since this topic is still not settled even in academia.

Use **Google Scholar / Google Scholar Labs** to search for relevant papers. Look for papers with a significant number of citations and papers coming from well-known companies, universities, and research institutions.

You are also free to use an LLM to understand topics or concepts that you are unfamiliar with.

The goal of this PS is not to test how well you can generate an answer using an LLM. It is to test **how well you can research, understand, compare, and form an opinion about an open research problem.**